$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$
$= 1 - 2\sin^2\alpha$
$= 2\cos^2\alpha - 1$

01. Tentukanlah nilai dari :

(a) $4.\cos^2 67{,}5^0 - 4\sin^2 67{,}5^0 + 6\sqrt{2}$

(b) $12\sqrt{3}\cos^2 15^0 - 6\sqrt{3}$

(a) $4\left(\cos^2 67{,}5^0 - \sin^2 67{,}5^0\right) + 6\sqrt{2}$

$4\left(\cos 2(67{,}5)\right) + 6\sqrt{2}$

$4\left(\cos 135\right) + 6\sqrt{2}$

$4\left(-\frac{1}{2}\sqrt{2}\right) + 6\sqrt{2}$

$-2\sqrt{2} + 6\sqrt{2}$

$4\sqrt{2}$ ✓

(b) $6\sqrt{3}\left(2\cos^2 15 - 1\right)$

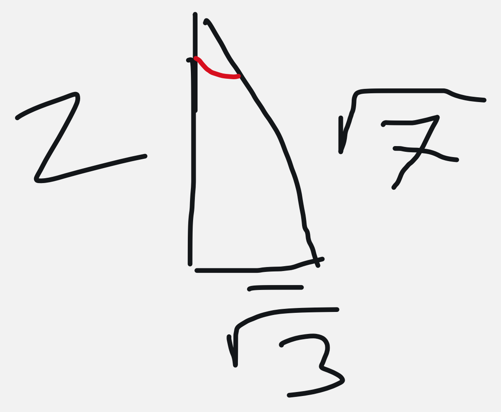
$6\sqrt{3}\left(\cos 2(15)\right)$

$\cancel{6}\sqrt{3} \cdot \frac{1}{\cancel{2}}\sqrt{3}$

$3\sqrt{9} = 3 \cdot 3$
$= 9$

Jika $\tan\alpha = \frac{1}{2}\sqrt{3}$ $= \sqrt{3}/2$ dan α sudut lancip, maka tentukanlah nilai $\sin 2\alpha$

$\sin 2\alpha = 2\sin\alpha\cos\alpha$



$\sin\alpha = \frac{\sqrt{3}}{\sqrt{7}}$

$\cos = \frac{2}{\sqrt{7}}$

$2\left(\frac{\sqrt{3}}{\sqrt{7}}\right)\left(\frac{2}{\sqrt{7}}\right)$

$2\left(\frac{2\sqrt{3}}{7}\right)$

$\frac{4\sqrt{3}}{7}$

Jika $\cos\alpha = -\frac{1}{\sqrt{3}}$ dan $90^0 < \alpha < 180^0$, maka tentukanlah nilai tan2α

$\sqrt{(\sqrt{3})^2 - 1} = \sqrt{3-1} = \sqrt{2}$

$\tan\alpha = \frac{\sqrt{2}}{1} = -\sqrt{2}$

$= \frac{2(-\sqrt{2})}{1-(-\sqrt{2})^2}$

$= \frac{-2\sqrt{2}}{1-2}$

$= 2\sqrt{2}$